different features [4]. That is, if `70M.L1.N111` fires, the model gets mixed signals that both the "prime **factors**" feature and the "International C**oven**" feature are present.

However, detecting n-grams in particular (and hence constructing the hypothesized pseudo-vocabulary), turns out to be a task particularly well suited for superposition (see A.6). In short, the model can take advantage of the fact that (for fixed n) exactly one possible n-gram out of $|\mathcal{V}|^n$ can occur at any time—that is, n-grams are mutually-exclusive binary features with respect to the current token of the input, despite coming from a massive set of possible features. While it would be impossible to dedicate a unique neuron to all possible n-grams, by leveraging the activations of multiple polysemantic neurons, each of which react to a bigram `XY` but with no other overlapping stimuli, the magnitude of the "true" feature gets boosted above all of the possible interfering features. As an example, in Figure 2(b), we show three neurons from the same layer which activate for the "social security" bigram while (mostly) not activating for bigrams with just one of the words (see the blue histograms being significantly to the right of the red and green histograms). However, because these neurons are polysemantic, there are many other inputs which cause them to activate, potentially even more so than the "social security" feature (see the orange histogram having a longer tail than the blue histograms). Despite this polysemanticity, by summing the activations across the three neurons, we achieve nearly perfect separation between the total activation magnitude of the "social security" feature and *all* other observed token combinations! We believe this to be first example of neuron superposition exhibited "in the wild" in real LLMs. We include examples for all 21 compounds words from layers 1-3 of Pythia 1B in Figures 9, 10, and 11, and further results on basis alignment in A.10 and Figures 13 and 14.

We believe that this computational motif—merging individual tokens into more semantically meaningful n-grams in the pseudo-vocabulary via a linear combination of massively polysemantic neurons—is one of the primary function of early layers. As a supporting line of evidence, we observe that a natural idealized model of n-gram recovery in superposition predicts a mechanistic fingerprint of superposition: the presence of large input weight norms and large negative input biases. We provide an idealised construction in Section C for how an arbitrary number of features can be compressed into two linear dimensions, similar to the construction in [4]. When we measure the product of the input weight norm and bias for each neuron in every layer and every model, we observe a striking difference in the early[6] layers, exactly in line with our conceptual argument (see Figure 2(c) for a summary and Figure 8 for the full distributions, and Section C for additional motivation of the metric and discussion).

We conclude by noting that, in another encouraging display of consilience, the phenomenon of different regions of the model employing different coding strategies for different functions bears resemblance to the diverse coding strategies employed by biological neural networks in different brain regions with different functional roles [74, 75, 76, 77]. We expect further research connecting local mechanisms to macroscopic structure to be particularly fruitful.

5.2 Context Neurons: a Monosemantic Neuron Family

Given the potential benefits of superposition, it is reasonable to ask if *any* features are represented monosemantically, and if so, why? We hypothesized that a likely candidate would be *context features*—high-level descriptions of all (or most) tokens within a sequence (e.g. `is_french` or `is_python_code`). Such features seem quite important, to the point that avoiding interference is worth a full neuron, while also being a higher-level property that may or may not be mutually exclusive or binary, making it harder to represent in superposition.

In particular, we probe on the language of natural language sequences from the Europarl dataset [78], the programming language of different Github source files, and the data subset of The Pile from which a sequence originated. We probe for these features using mean aggregation—training a classifier on the averaged activations over each sequence to predict the sequence label (e.g., `is_french` or `is_python`). Figure 3 depicts a sample of our results, illustrating the existence of highly specialized context neurons that activate approximately only when a token is in a specific context[7] (with 21 additional examples in Figures 15, 16, and 17). To better understand the function of these neurons, for each token, we study the distribution of activations broken down by context (e.g., the activation of "return" in every programming language). Sorting by the largest differences (middle panel), we observe that one explanation for these neurons' roles is token disambiguation. For example, many programming languages have a `return` keyword, but neuron `1B.L6.N3108` activates on `return` if and only if it is in the context of Go code.

To further support our interpretation and gain more insight into function, we conduct ablation experiments comparing the language modeling loss of the base model to the loss of the model with the identified neuron fixed to 0 for all tokens in every sequence (right panel). As anticipated, ablating the context neurons significantly degrades language modeling performance within the relevant context, while leaving other contexts nearly unaffected. The impact, however, heavily

[6] Pythia models use parallel attention so layer 0 is purely a function of the current token, hence not subject to the n-gram analysis.

[7] By recording the activation of every token, we eliminate the concern of finding an `is_french_noun` neuron instead of an `is_french` neuron.

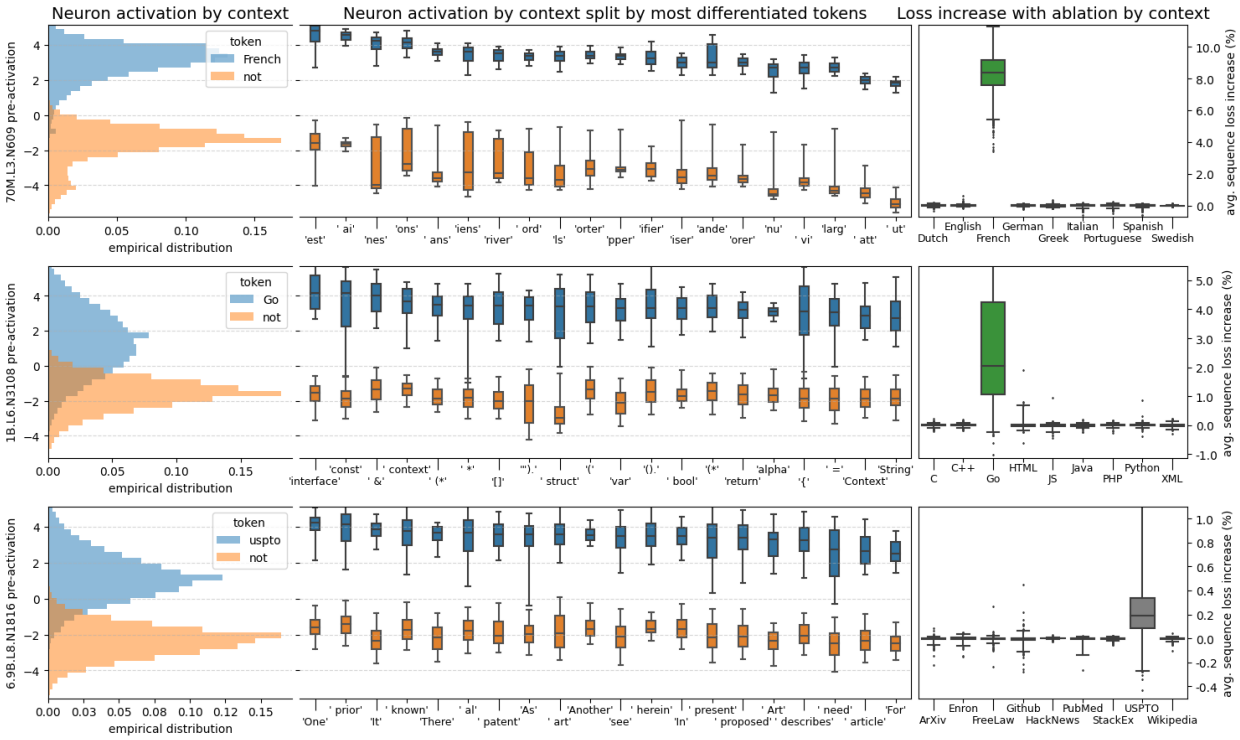


Figure 3: Analysis of individual context neurons that activate on tokens in French (top), in Go code (middle), and US Patent office documents (bottom). We show the distribution of activation when in and not in the target context (left) in addition to the tokens with average activation most distinguished by the neuron (middle). Finally we report the increase in sequence loss across contexts when the neuron is ablated (right).

depends on the model size—in the 70M parameter model ($\approx$ 12k neurons), ablating a single neuron causes an average loss increase of 8% per French sequence, while in the 6.9B model ($\approx$524k neurons), ablating one neuron results in only a 0.2% increase in loss.

While these neurons appear to be genuinely monosemantic, we emphasize that it is extremely difficult to prove this. Doing so requires answering thorny ontological questions (what *is* French?) and then efficiently searching through a dataset of billions of tokens to verify the neuron implements the answer. Moreover, the "true" feature the neuron responds to might be quite subtle. For instance, many of the code neurons do not fire when in a code comment, whereas the French neuron will fire in a non-French context on a French name or other token strongly associated to France[8]. In some sense, these exceptions prove the rule of the primary role of the neuron, but highlight the difficulty of mapping human features to the ontology of the network. Perhaps most challenging though is ruling out the existence of very rare features the neuron also responds to. Sufficiently rare features will likely be undetectable from random samples or summary statistics—even when broken down by token—and therefore require manually[9] examining max activating dataset examples in subdistributions where we don't expect to find much French text.

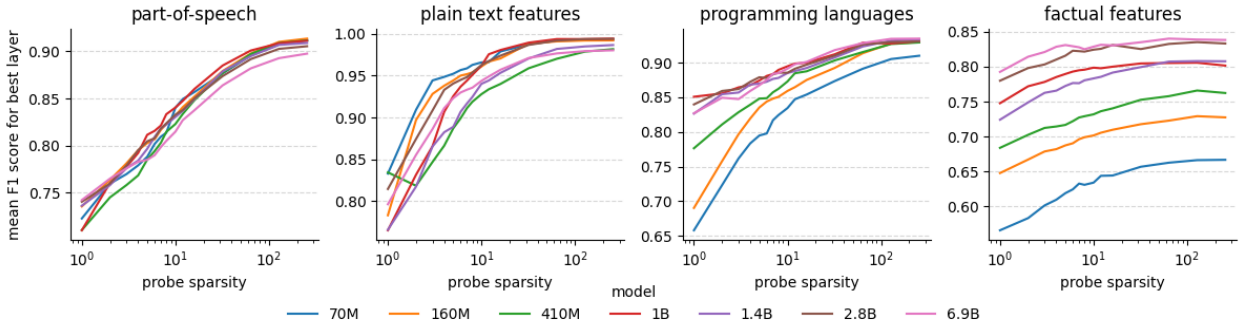
5.3 Effects of Scale: Quantization and Splitting

Given the importance of scale in LLMs, we turn our attention to studying the relationship between model size and the sparsity of representations, and the dynamics that drive this relationship. For all of our feature datasets described in B.2, we train a series of probes sweeping the value of k from 256 to 1 using adaptive thresholding. For sake of summarization, we report the maximum out-of-sample F1 value over the layers for each model, while averaging over the features within the collection (see Figure 4(a) for a sample and Figure 7 for full results with random baselines).

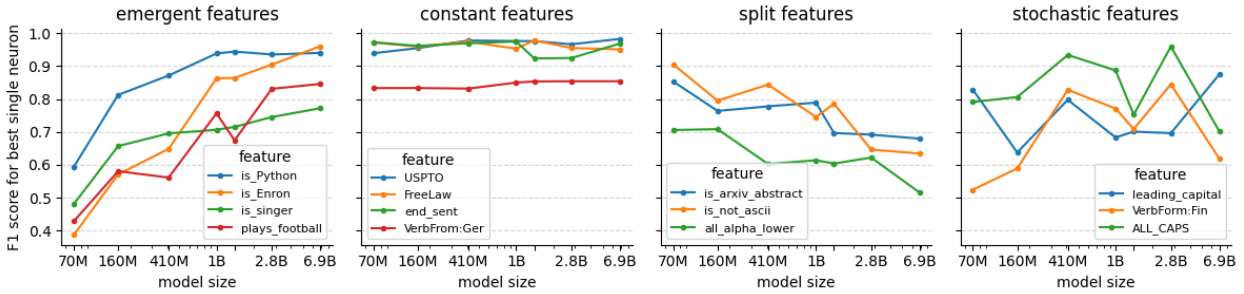
While some features appear to be more sparsely represented with scale (e.g., programming languages and factual features), we were surprised by how consistent others were (e.g., part-of-speech and compound words). In fact, for some

[8]To further explore this, we encourage the reader to explore max activating dataset examples in Neuroscope[73]

[9]This is an area where we expect AI assisted interpretability to be particularly useful.



(a) Probe sparsity versus classification performance for different feature types and model sizes.



(b) Classification performance of most feature-aligned neuron by model size for different feature classes.

Figure 4: Summary of results on model scale. Representational sparsity increases on average with scale (top) but individual features obey different scaling dynamics (bottom).

of the simplest plain-text features, the smallest models seem to actually implement greater sparsity. When analyzing individual features on a per neuron basis, several general patterns become more apparent (Figure 4(b)). We believe there are two main dynamics driving these results: the quantization model of scaling [79] and neuron splitting [10].

In particular, the quantization hypothesis posits that there exists a natural ordering of features learned by a model with increasing scale based on how loss reducing they are, with larger models able to learn a longer tail of increasingly rare features [79]. In our context, this suggests that features like part-of-speech or compound words are within the feature set learned by even small models, and are represented with very similar sparsity patterns. In contrast, factual features and some (but not all) contextual features only get represented by single neurons at sufficient scales. As a countervailing force, with increasing scale the model can dedicate multiple neurons to represent more granular features that previously comprised one coarser feature. Consider the `ALL_CAPS` feature; while it would likely be advantageous to have dedicated circuitry for representing this feature in all models, a larger model might have dedicated neurons for all of the particular reasons a token might be in all capital letters (e.g., an abbreviation, a constant in python, shouting on the internet, etc.), eliminating the need to have just one more coarse grained representation. The result, as viewed from a probing experiment, is less sparsity, not more.

5.4 Refining and Verifying Interpretations

The result of a sparse probing experiment is a set of k neurons with a collection of classification performance metrics, usually for a range of values of k. We illustrate what conclusions might follow, simple techniques to refine and corroborate such conclusions, and confounding factors to be wary of.

One confounding factor is an ambiguity between superposition and *composition* [12]. If a (layer, feature) pair has a 1-sparse probe with poor accuracy and a k-sparse probe with high accuracy, it is temping to take this as evidence of superposition, but it is also consistent with a feature being coarse or compositional—being either the union or intersection of multiple independent features. As an example, consider an `is_athlete` feature. Such a feature could either be represented by a single neuron, a superposition of polysemantic neurons, or as the union of neurons for specific types of athletes, as those shown in Figure 5(a). While we can distinguish a feature union from superposition by analyzing the activation co-occurrence (only one athlete type neuron activates for each athlete), this reasoning does not work for a true compositional feature. For example, we also found individual neurons which coded for